MULTIPLE LINEAR REGRESSION MODEL:

- A Type of regression model in which we have at least two independent variables are called multiple linear model. E.g $Y = \alpha + \beta_1 X_1 + \beta_2 X_2 + \epsilon$
  Where α , β_1 & β_2 are parameters of regression model which will be estimated by using ordinary least square.
- E.g income depends on number of employees, floor area etc
- Bp depends on age, weight, diet, exercise etc
- The yield of crop depends on the amount of fertilizer, amount of land, rain fall, quality of seeds etc

Now its estimated model is $\hat{Y} = a + b_1 X_1 + b_2 X_2$

$$a = \bar{Y} - b_1\overline{X_1} - b_2\overline{X_2} \quad (\bar{Y} = \frac{\sum Y}{n}\ , \bar{X}_1 = \frac{\sum X_1}{n}.\ \bar{X}_2 = \frac{\sum X_2}{n})$$

$$b_1 = \frac{\sum x_1 y \sum x_2^2 - \sum x_2 y \sum x_1 x_2}{\sum x_1^2 \sum x_2^2 - (\sum x_1 x_2)^2}$$

$$b_2 = \frac{\sum x_2 y \sum x_1^2 - \sum x_1 y \sum x_1 x_2}{\sum x_1^2 \sum x_2^2 - (\sum x_1 x_2)^2}$$

$$\sum x_1^2 = \sum X_1^2 - \frac{(\sum X_1)^2}{n_1}$$

$$\sum x_2^2 = \sum X_2^2 - \frac{(\sum X_2)^2}{n_2}$$

$$\sum x_1 x_2 = \sum X_1 X_2 - \frac{(\sum X_1)(\sum X_2)}{n}$$

$$\sum x_1 y = \sum X_1 Y - \frac{(\sum X_1)(\sum Y)}{n}$$

$$\sum x_2 y = \sum X_2 Y - \frac{(\sum X_2)(\sum Y)}{n}$$

Example) A statistician wants to predict the income of restaurants, using two independent variables: the number of restaurant employees and restaurant floor area. He collected the following data:

Income (000) (Y)	Floor area (Square feet) (X1)	# of Employees (X2)
30	10	15
22	05	08
16	10	12
07	03	07
14	02	10

Fit the multiple linear regression model $\hat{Y} = a + b_1X_1 + b_2X_2$. Interpret your model. Also estimate income(Y) when floor area(X1) is 15 square feet and number of employees(X2) are 20.

Ans:

Y	X_1	X_2	${X_1}^2$	${X_2}^2$	X_1X_2	X_1Y	X_2Y	Y^2
30	10	15	10*10 =100	15*15=225	10*15=150	30*10 =300	30*15 = 450	900
22	05	08	5*5 = 25	8*8 = 64	5*8 = 40	22*5 =110	22*8 = 176	484
16	10	12	10*10 =100	12*12 =144	10*12= 120	16*10 =160	16*12 =192	256
07	03	07	3*3 = 09	7*7 = 49	3*7 = 21	7*3 = 21	7*7 = 49	49
14	02	10	2*2 = 04	10*10 =100	2*10 = 20	14*2 = 28	14*10 =140	196
$\sum Y = 89$	$\sum X_1$ =30	$\sum X_2$ =52	$\sum {X_1}^2$ =238	$\sum {X_2}^2$ =582	$\sum X_1X_2$ =351	$\sum X_1Y$ =619	$\sum X_2Y$ =1007	$\sum Y^2$ =1885

Now multiple linear regression model income(Y) on floor area(X1) and # of employee (X2) is

$$\hat{Y} = a + b_1X_1 + b_2X_2$$

$$\sum {x_1}^2 = \sum {X_1}^2 - \frac{(\sum X_1)^2}{n}$$

n = 5, $\sum {X_1}^2$ =238, $\sum X_1$ =30

$$\sum {x_1}^2 = 238 - \frac{(30)^2}{5} = 58$$

$$\sum x_2{}^2 = \sum X_2{}^2 - \frac{(\sum X_2)^2}{n}$$

$$\sum x_2{}^2 = 582 - \frac{(52)^2}{5} = 41.2$$

$$\sum x_1 x_2 = \sum X_1 X_2 - \frac{(\sum X_2)(\sum X_1)}{n}$$

$$\sum x_1 x_2 = 351 - \frac{(30)(52)}{5} = 39$$

$$\sum x_1 y = \sum X_1 Y - \frac{(\sum Y)(\sum X_1)}{n}$$

$$\sum x_1 y = 619 - \frac{(30)(89)}{5} = 85$$

$$\sum x_2 y = \sum X_2 Y - \frac{(\sum Y)(\sum X_2)}{n}$$

$$\sum x_2 y = 1007 - \frac{(52)(89)}{5} = 81.4$$

$$b_1 = \frac{\sum x_1 y \sum x_2^2 \quad - \sum x_2 y \sum x_1 x_2}{\sum x_1^2 \sum x_2^2 - (\sum x_1 x_2)^2}$$

$$b_1 = \frac{(85)(41.2) - (81.4)(39)}{(58)(41.2) - (39)^2} = \frac{327.4}{868.6} = 0.38$$

Similarly $b_2 = \frac{\sum x_2 y \sum x_1^2 \quad - \sum x_1 y \sum x_1 x_2}{\sum x_1^2 \sum x_2^2 - (\sum x_1 x_2)^2}$

$$b_2 = \frac{(81.4)(58) - (85)(39)}{(58)(41.2) - (39)^2} = \frac{1406.2}{868.6} = 1.62$$

$$a = \bar{Y} - b_1 \overline{X_1} - b_2 \overline{X_2} \quad (\bar{Y} = \frac{\sum Y}{n} = \frac{89}{5} = 17.8 \;, \bar{X}_1 = \frac{\sum X_1}{n} = \frac{30}{5} = 06. \;\; \bar{X}_2 = \frac{\sum X_2}{n} = \frac{52}{5} = 10.4)$$

$$a = 17.8 - (0.38)(6) - (1.62)(10.4) = -1.33$$

Now fitted regression line income (Y) on floor area(X1) and number of employee (X2) is

$$\hat{Y} = a + b_1 X_1 + b_2 X_2$$

$$\hat{Y}(Income) = -1.33 + 0.38 X_1 (Floor\ Area) + 1.62 X_2 (\#\ of\ employees)$$

Interpretation of the model:

- If there is no floor area (X1=0) and there is no employee (X2), then income (Y) will be decrease by Rs. 1330/-.

- If there is unit square foot floor area (X1=0) and there is no employee (X2=0), the income (Y) will be increased by Rs. 380/-
- If there is floor area (X1=0) and unit number of employee (X2=1), the income (Y) will be increased by Rs. 1620/-
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Estimation:

$$\hat{Y}(Income) = -1.33 + 0.38X_1(Floor\ area\ in\ square\ feet) + 1.62X_2(\#\ of\ employees)$$

Put X1 =15, X2 =20

$$\hat{Y}(Income) = -1.33 + 0.38(15) + 1.62(20)$$
$$\hat{Y}(Income) = -1.33 + 5.7 + 32.4$$
$$\hat{Y}(Income) = 36.77$$
$$\hat{Y}(Income) = Rs. 36770/-$$

Coefficient of Determination

- It shows how much changes in dependent variable due to independent variable./it explained how much changes in dependent variable due to independent variable.
- It is denoted byR^2 and obtained as

$$R^2{}_{Y.12} = \frac{explained\ variation}{Total\ variation}$$

$$R^2{}_{Y.12} = \frac{\sum(\hat{Y} - \bar{Y})^2}{\sum(Y - \bar{Y})^2}$$

OR

$$R^2{}_{Y.12} = 1 - \frac{Unexplained\ variation}{Total\ variation}$$

$$R^2{}_{Y.12} = 1 - \frac{\sum(Y - \hat{Y})^2}{\sum(Y - \bar{Y})^2}$$

- $\sum\left(Y - \hat{Y}\right)^2 = \sum Y^2 - a\sum Y - b_1\sum X_1Y - b_2\sum X_2Y$
- $\sum(Y - \bar{Y})^2 = \sum Y^2 - \frac{(\sum Y)^2}{n}$

- Range of coefficient of determination lies b/w 0 to +1.
- Coefficient of determination is square of correlation coefficient.

Example) Data from above question

n = 5, $\sum Y = 89$, $\sum X_1$ =30 , $\sum X_2$ =52 , $\sum X_1{}^2$ =238 , $\sum X_2{}^2$ =582 , $\sum X_1X_2$ =351 , $\sum X_1Y$ =619, $\sum X_2Y$ =1007, $\sum Y^2 = 1885$, $a = -1.33$, $b_1 = 0.38$, $b_2 = 1.62$

$$R^2{}_{Y.12} = 1 - \frac{\sum(Y - \hat{Y})^2}{\sum(Y - \bar{Y})^2}$$

$$\sum\left(Y - \hat{Y}\right)^2 = \sum Y^2 - a\sum Y - b_1\sum X_1Y - b_2\sum X_2Y$$

$$\sum\left(Y - \hat{Y}\right)^2 = 1885 - (-1.33)(89) - (0.38)(619) - (1.62)(1007)$$

$$\sum\left(Y - \hat{Y}\right)^2 = 1885 + 118.37 - 235.22 - 1631.34 = 136.81$$

$$\sum(Y - \bar{Y})^2 = \sum Y^2 - \frac{(\sum Y)^2}{n} = 1885 - \frac{(89)^2}{5} = 1885 - \frac{7921}{5}$$

$$\sum(Y - \bar{Y})^2 = 1885 - 1584.2 = 300.80$$

$$R^2{}_{Y.12} = 1 - \frac{\sum\left(Y - \hat{Y}\right)^2}{\sum(Y - \bar{Y})^2} = 1 - \frac{136.81}{300.80} = 1 - 0.454 = 0.55$$

Interpretation of R^2:

- $R^2 = 0.55$ shows that 55% changes in income (Y) is due to floor area (X1) and number of employees (X2) and 45% is due to other factors which is not included in the model.

Error of the Model: $Error\ of\ the\ model = \sum(Y - \hat{Y})$

Y	X_1	X_2	$\hat{Y} = -1.33 + 0.38X_1 + 1.62X_2$	$e = (Y - \hat{Y})$
30	10	15	$\hat{Y} = -1.33 + 0.38(10) + 1.62(15) = 26.77$	30 – 26.77 = 3.23
22	05	08	$\hat{Y} = -1.33 + 0.38(5) + 1.62(8) = 13.53$	22 – 13.53 = 8.47
16	10	12	$\hat{Y} = -1.33 + 0.38(10) + 1.62(12) = 21.91$	16 – 21 .91 = -5.91
07	03	07	$\hat{Y} = -1.33 + 0.38(3) + 1.62(7) = 11.15$	7 – 11.15 = - 4.15
14	02	10	$\hat{Y} = -1.33 + 0.38(2) + 1.62(10) = 15.63$	14 – 15.63 = -1.63
				$\sum e = 0.01$